(i)

$$G_{R_1}(x) = \frac{1}{6}(x^0 + x + x^2 + x^3 + x^4 + x^5)$$

$$\begin{aligned} G_{R_2}(x) &= \frac{1}{36}(x^0 + x + x^2 + x^3 + x^4 + x^5)^2 \\ &= \frac{6}{36}(x^0 + x + x^2 + x^3 + x^4 + x^5) \\ &= G(x) \end{aligned}$$

The coefficient of x^0 is $\frac{1}{6}$, which is the probability that S_n is divisible by 6.

(ii)

$$G_1(x) = \frac{1}{6}(1 + 2x + x^2 + x^3 + x^4)$$

$$G_1(x) = \frac{1}{6}(x + y)$$

Modulo $x^5 = 1$, we have

$$xy = x(1 + x + x^2 + x^3 + x^4) = x + x^2 + x^3 + x^4 + x^5 = y, \qquad y^2 = 5y$$

More generally, $x^m y = y$ for all integers $m \geq 1$.

$$\begin{aligned} G_2(x) &= \frac{1}{36}(x + y)^2 \\ &= \frac{1}{36}(x^2 + 2xy + y^2) \\ &= \boxed{\frac{1}{36}(x^2 + 7y)} \end{aligned}$$

$$\begin{aligned} G_3(x) &= \frac{1}{6^3}(x + y)(x^2 + 7y) \\ &= \frac{1}{6^3}(x^3 + x^2 y + 7xy + 7y^2) \\ &= \frac{1}{6^3}(x^3 + y + 7y + 35y) \\ &= \frac{1}{6^3}(x^3 + 43y) \end{aligned}$$

$$\begin{aligned} G_4(x) &= \frac{1}{6^4}(x + y)(x^3 + 43y) \\ &= \frac{1}{6^4}(x^4 + x^3 y + 43xy + 43y^2) \\ &= \frac{1}{6^4}(x^4 + y + 43y + 215y) \\ &= \frac{1}{6^4}(x^4 + 259y) \end{aligned}$$

From $G_1(x), G_2(x), G_3(x), G_4(x)$, assume

$$G_k(x) = \frac{1}{6^k}\left(x^k + \frac{6^k - 1}{5}y\right)$$

Base case $n = 1$: $\quad G_1(x) = \dfrac{1}{6}(x + y)$

$$\begin{aligned} G_{k+1}(x) &= \frac{1}{6^{k+1}}\left(x^k + \frac{6^k - 1}{5}y\right)(x + y) \\ &= \frac{1}{6^{k+1}}\left(x^{k+1} + x^k y + \frac{6^k - 1}{5}xy + \frac{6^k - 1}{5}y^2\right) \\ &= \frac{1}{6^{k+1}}\left(x^{k+1} + \left[1 + 6\left(\frac{6^k - 1}{5}\right)\right]y\right) \\ &= \frac{1}{6^{k+1}}\left(x^{k+1} + \frac{6^{k+1} - 1}{5}y\right) \end{aligned}$$

Due to mathematical induction, the statement is true for all $n \geq 1$.
If $5 \nmid n$, then x^n contributes no x^0 term modulo $x^5 = 1$,

$$\mathrm{P}(5 \mid S_n) = \frac{1}{6^n} \cdot \frac{6^n - 1}{5} = \boxed{\frac{1}{5}\left(1 - \frac{1}{6^n}\right)}$$

If $5 \mid n$, then $x^n = x^0$ modulo $x^5 = 1$,

$$\mathrm{P}(5 \mid S_n) = \frac{1}{6^n}\left(1 + \frac{6^n - 1}{5}\right) = \boxed{\frac{1}{5}\left(1 + \frac{4}{6^n}\right)}$$

This solution was submitted by **Jon**
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